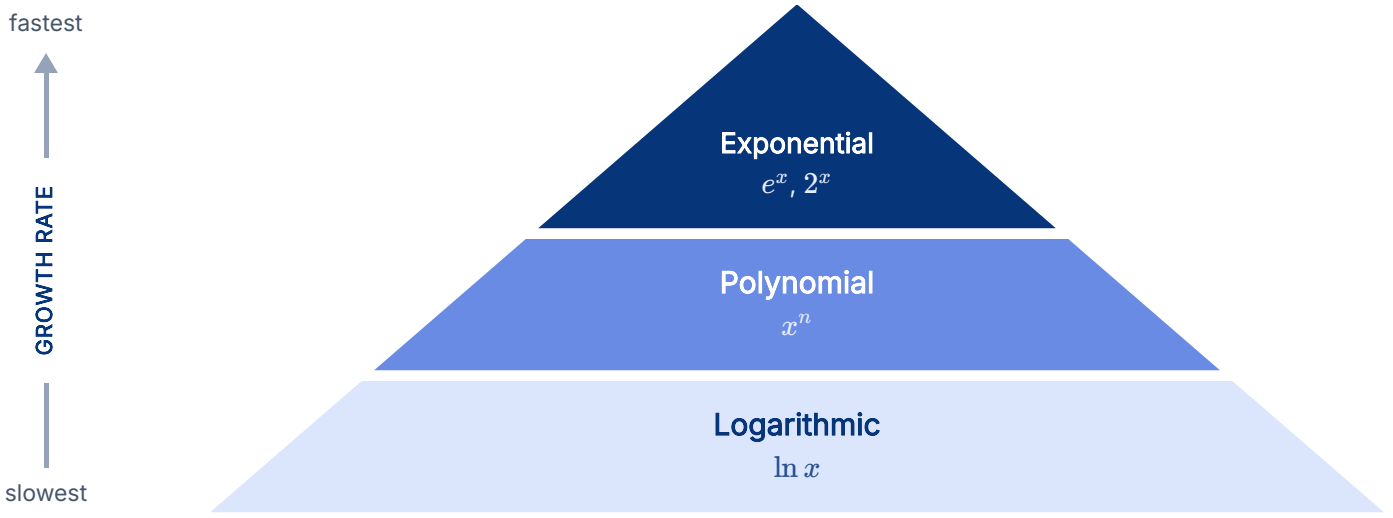
BEFORE YOU READ

Who wins as $x \to \infty$

Different function types grow at fundamentally different rates. Each level eventually overtakes every function on the levels below it.



TAKEAWAY

What the rank buys

Any exponential beats any polynomial, $\lim_{x \to \infty} \frac{x^n}{e^x} = 0$, and any polynomial beats any logarithm, $\lim_{x \to \infty} \frac{\ln x}{x^n} = 0$ for $n > 0$.